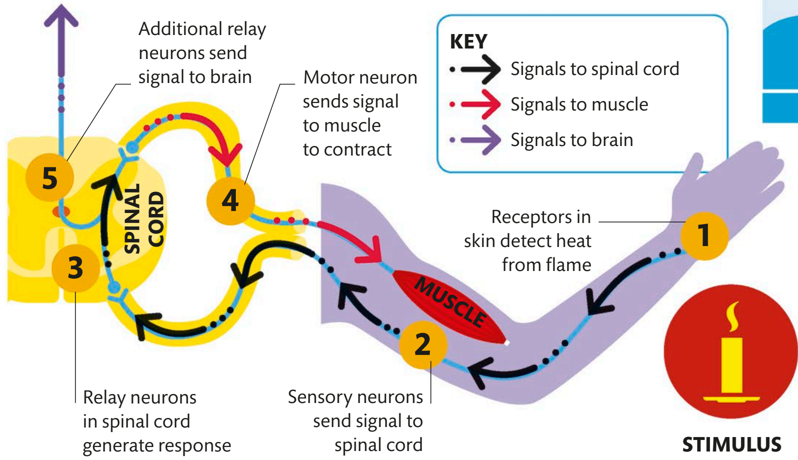
Reflex actions

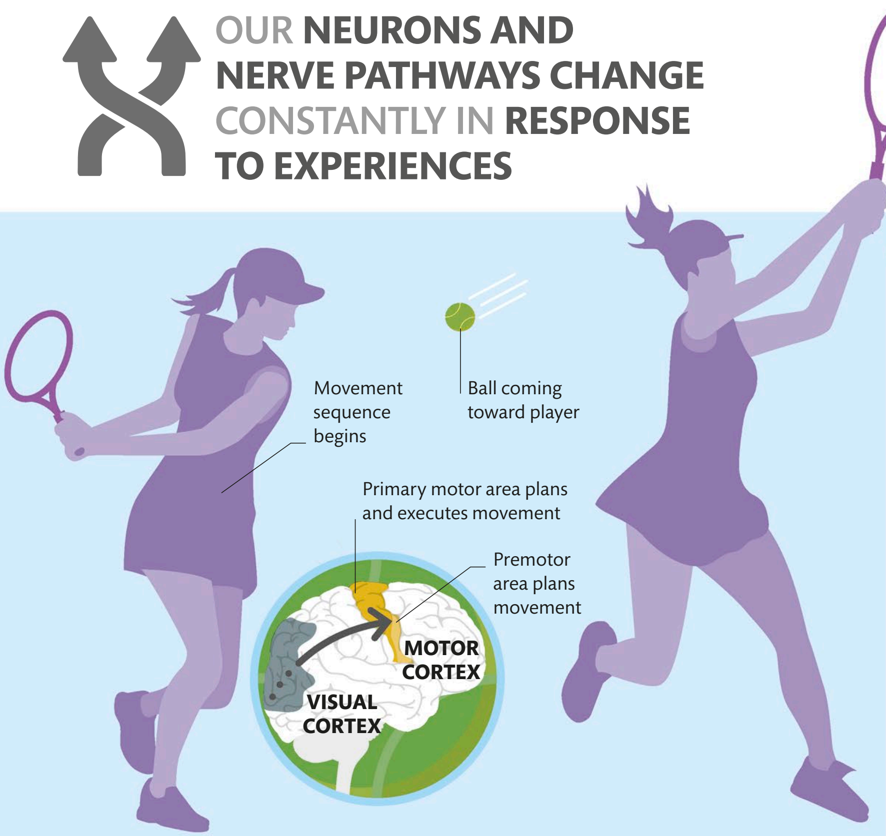
Reflexes are split-second responses to danger that we do not have to learn or even think about; the body reacts automatically. Reflex actions involve the same muscles that are used in voluntary movements, but the initial, instantaneous response does not involve the brain. Instead, the signal from the sensory nerves travels to the spinal cord, which triggers a response that travels along the motor nerves. Additional signals are sent to the brain afterward, to encode the memory in case the danger recurs.



OUR NEURONS AND NERVE PATHWAYS CHANGE CONSTANTLY IN RESPONSE TO EXPERIENCES

Bypassing the brain

Reflexes involve a simple neural response called the reflex arc. Receptors in the skin and muscles send a danger signal along sensory neurons to the spinal cord; there, relay neurons synapse with motor neurons to trigger a fast response.



3 Planning
The brain combines real-time visual information and stored programs for movement sequences to create a plan of action. This is first rehearsed in the premotor area and then sent to the primary motor cortex.

4 Conscious action
By the time the player becomes conscious of acting, the movement sequence is well underway. The action is most likely to be effective if the person has sufficient skill, stored knowledge, and information.

DEVELOPING COMPETENCE

Anyone learning a new skill passes through several stages. Beginners have to work hard to acquire competence. With practice, neural pathways develop until the learner can perform well without thinking about it.